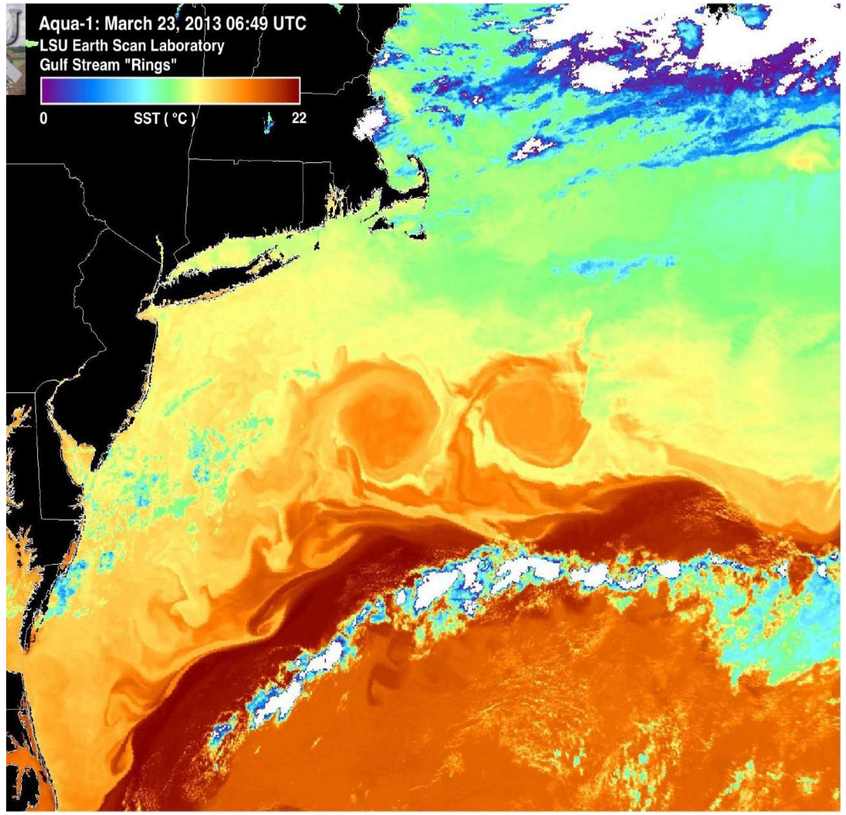
IRC-MABL-ASI Workshop
Seoul National University, June 22-23, 2026

Directional Wave–Wind Coupling, Drag Coefficient Rolloff, and Air-Sea Fluxes Under Tropical Cyclones

Hyodae Seo
University of Hawai‘i at Mānoa

UH: César Sauvage, Wyn Pauly, Sachiko Yoshida, Ben Barr
OIST: Satoshi Mitarai
WHOI: Carol Anne Clayson, Jake Davis

Air-Sea Interaction







UNIVERSITY of HAWAI'I® MĀNOA

https://www.HyodaeSeo.com

WOODS HOLE OCEANOGRAPHIC INSTITUTION

Drag Coefficient and Storm Potential Intensity

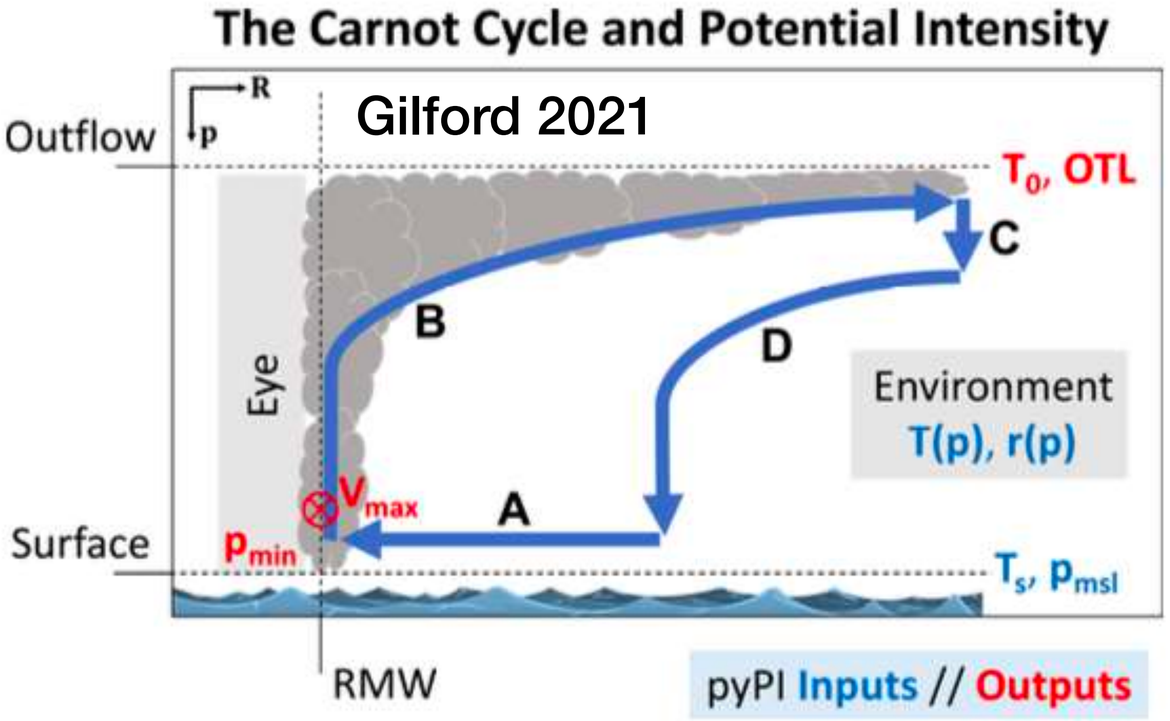
In the equilbirum condtion, TC intensity is contrained by the balance between energy added at the warm ocean surface and energy lost in the cool outflow region aloft (Emanuel 1988)

$$V_{max}^2 = \frac{(\overline{T} - T_o)}{T_o} \frac{C_K}{C_D} (K_{SST} - k)$$

TC outflow tempearture

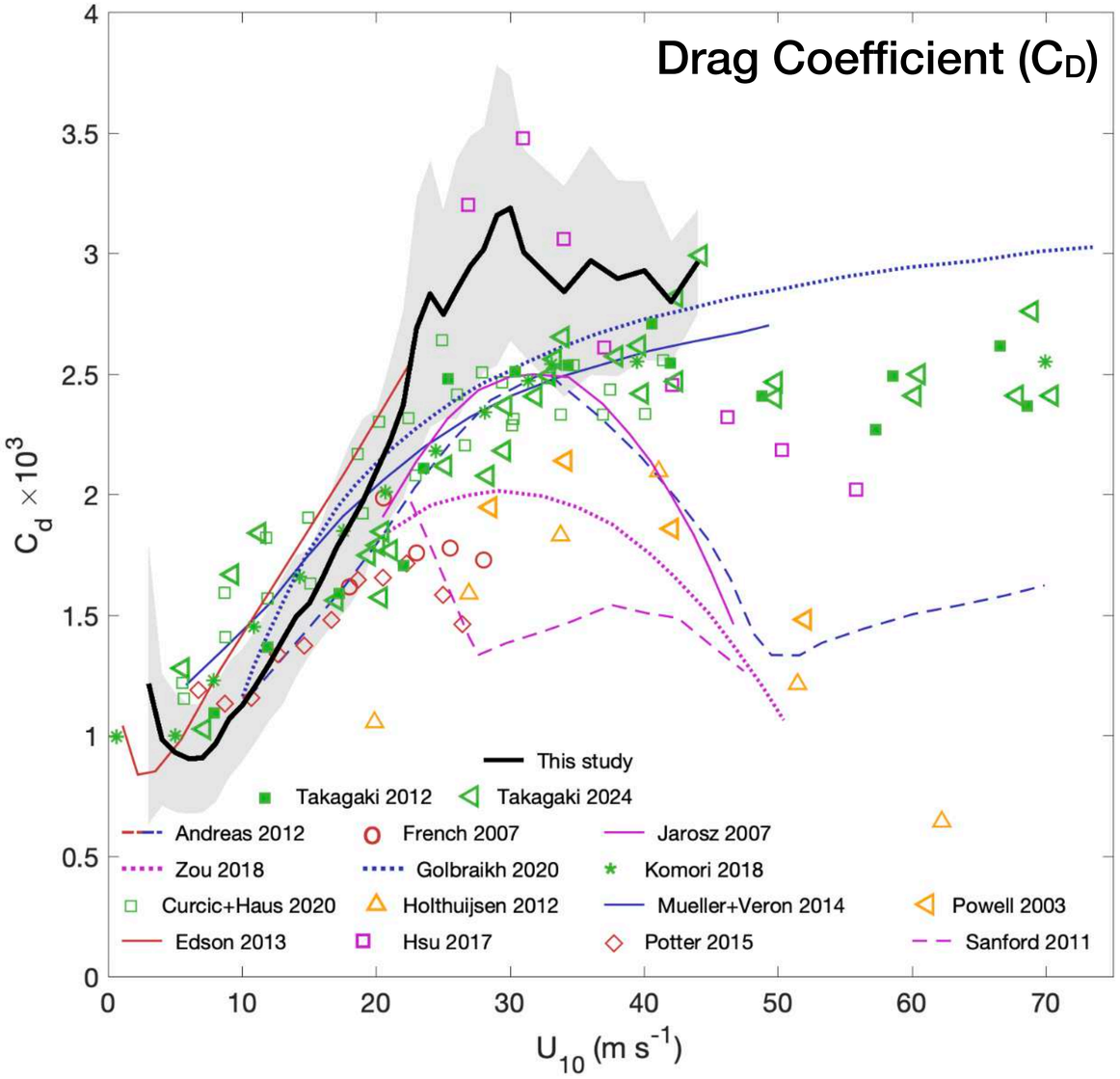
pre-storm ocean temp.

air-sea enthalpy difference



C_K, C_D, k, & K_{SST} all vary with sea state and feature strong TC spatial variability.
Sea states and wave-supported roughness length are key parameters.

Sea state and drag coefficient under very high winds





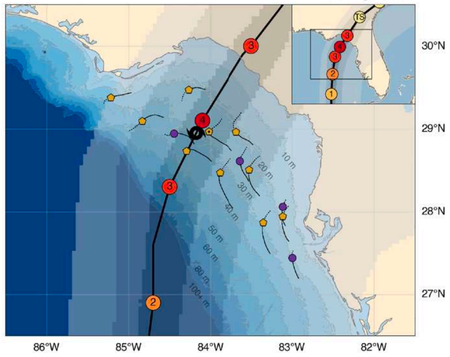
https://www.youtube.com/results?search_query=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3DTzgdSSb0T98

The hurricane foam coverage and sea spray form a slippery surface: Wind energy input to waves ceases to increase while wind speeds increase

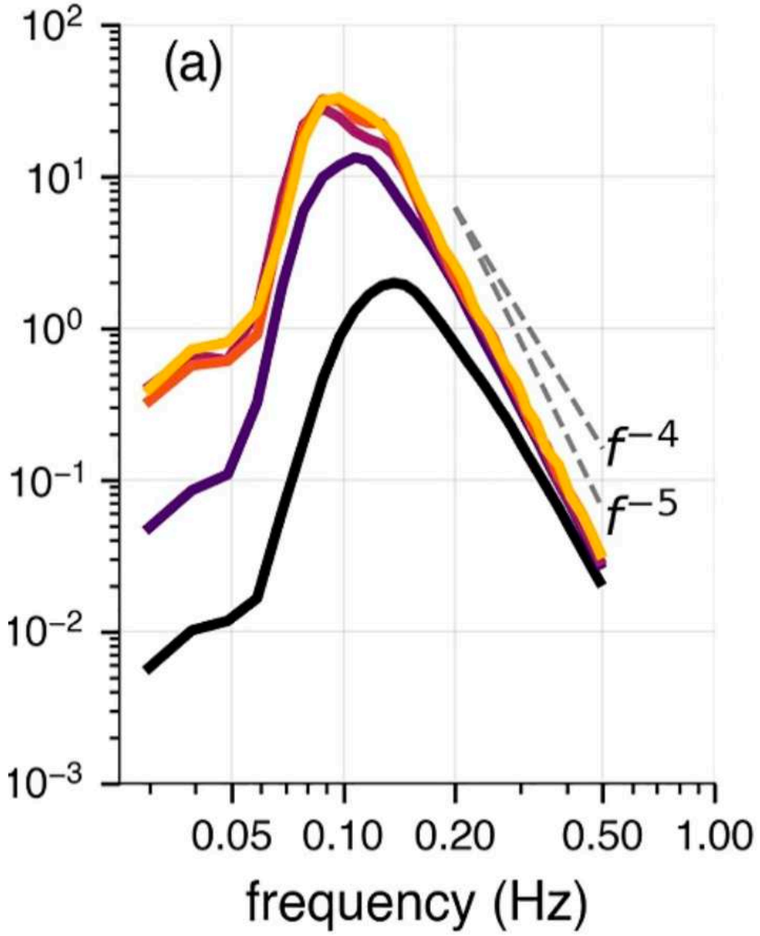
Foltz et al. 2026

Mean square slope (MSS) and Hurricane Idalia (2023)

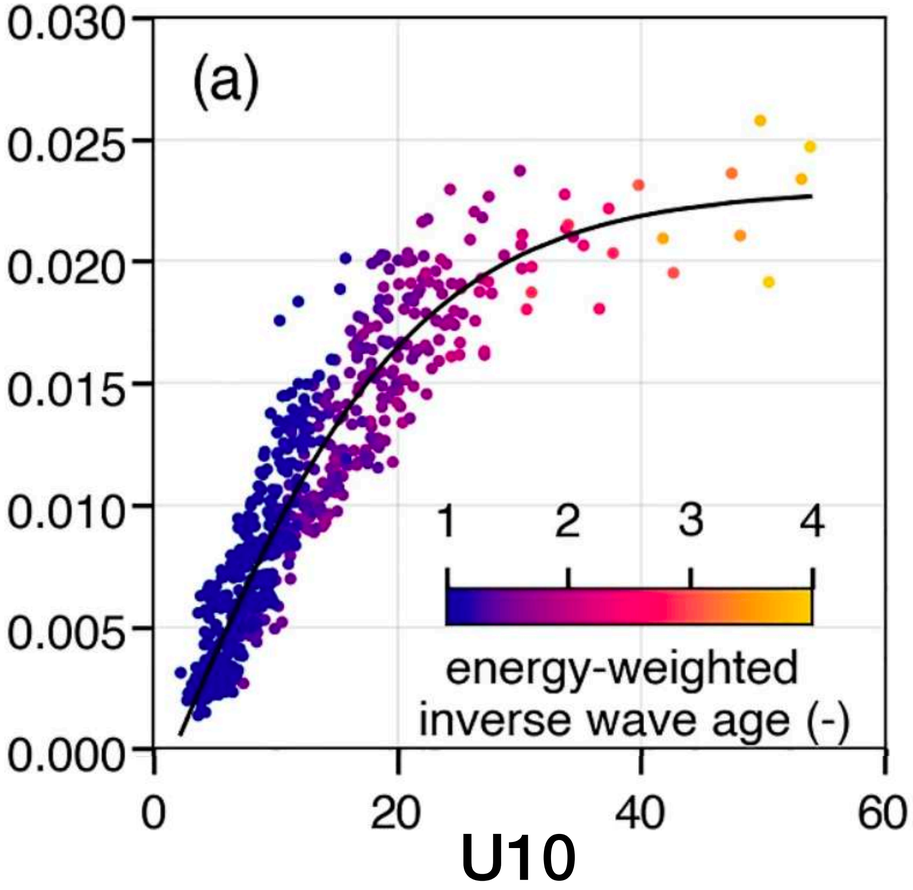




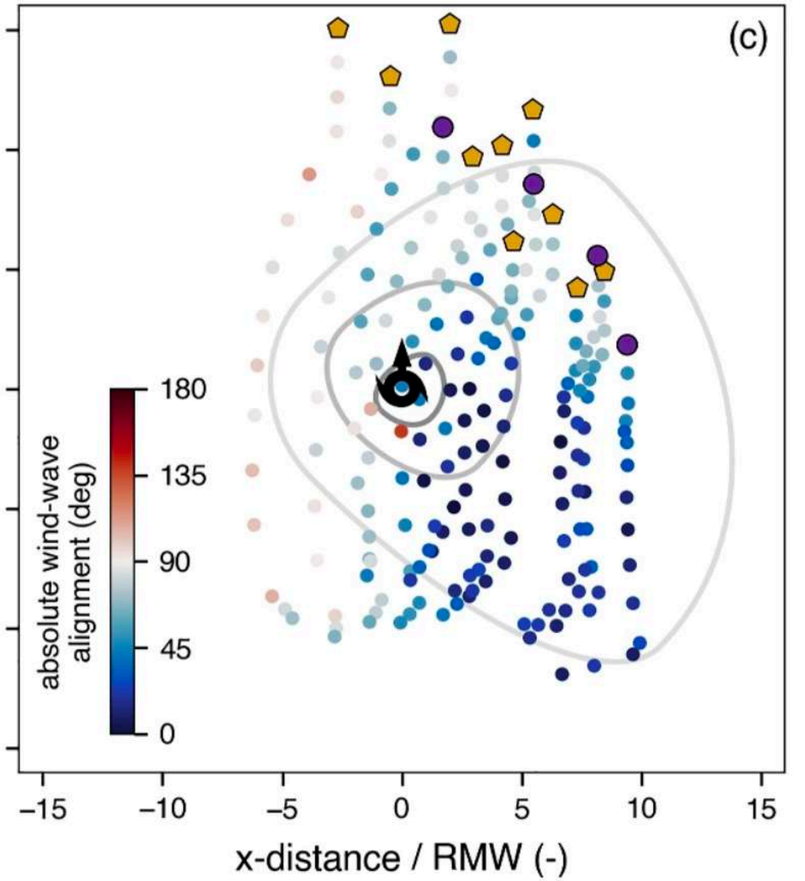
Mean energy density



MSS $\int_{f_1}^{f_2} \frac{(2\pi f)^4 E(f)}{g^2} df$

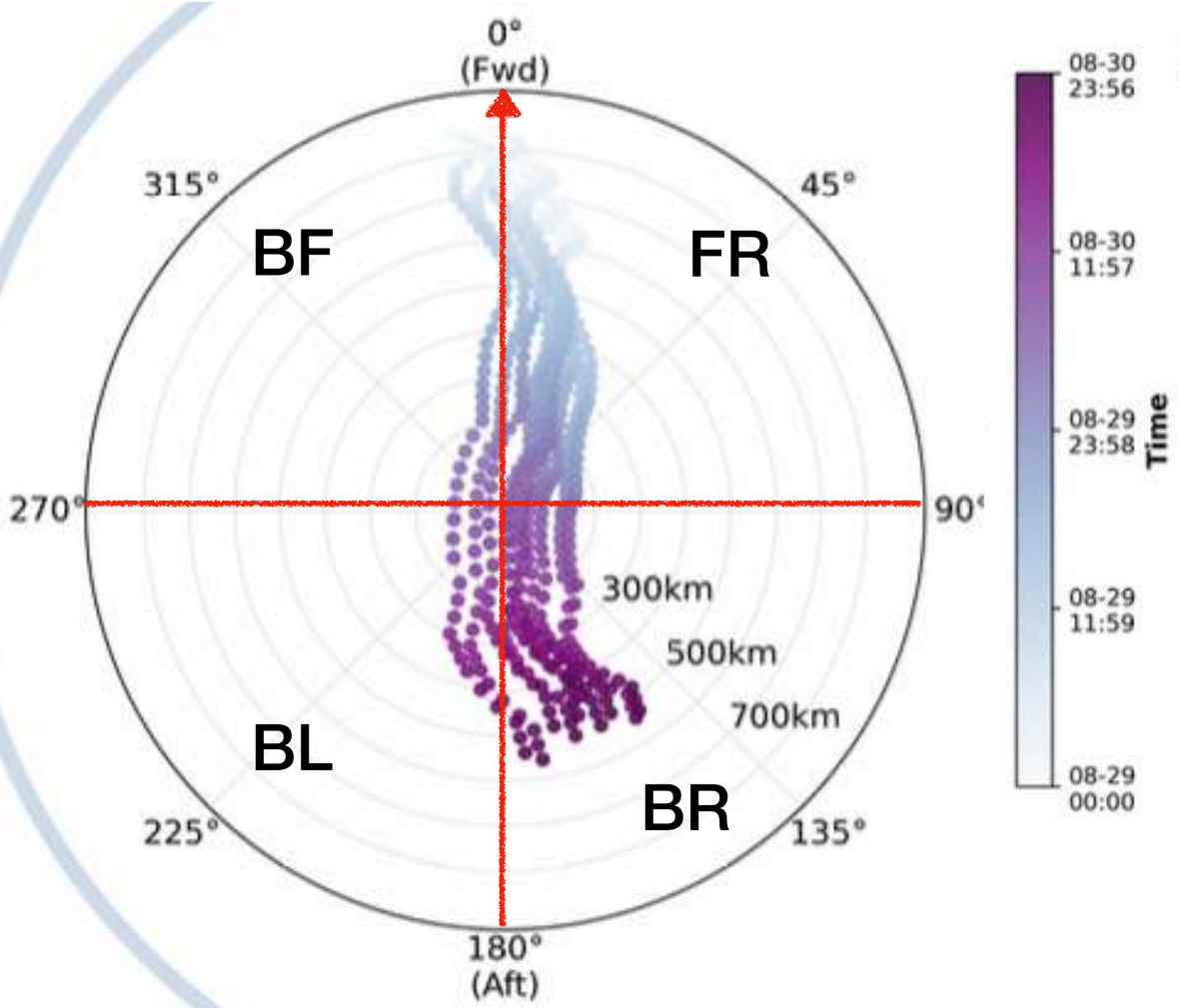


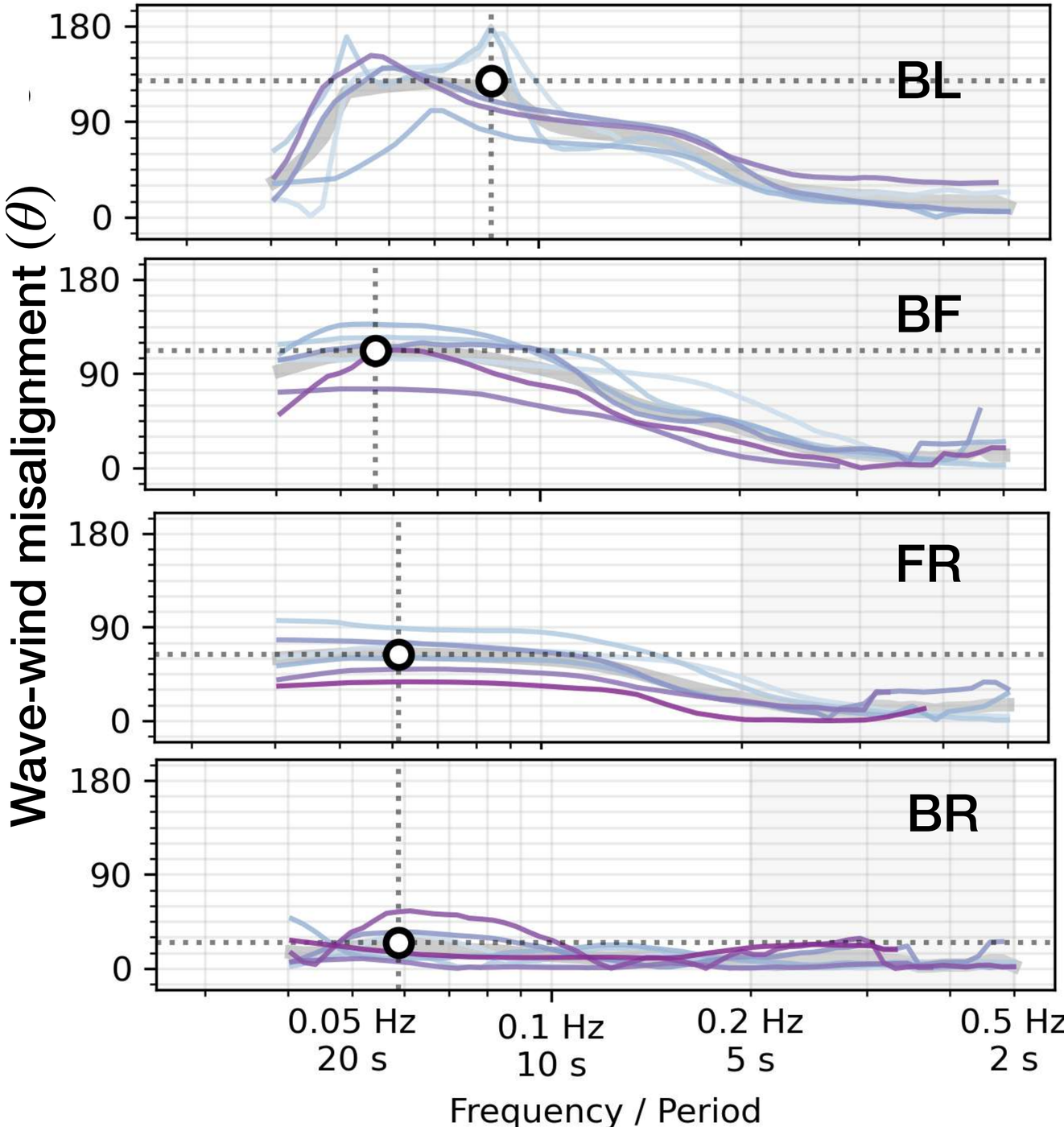
Wave-wind misalignment (θ)



Davis et al. (2025)

Wave-wind mislaignment and Hurricane Idalia (2023)





Wyn Pauly

COARE bulk flux parameterization (Edson et al. 2013)

$$\tau = \rho_a C_D U_r^2 \qquad C_D \cong \left[\frac{\kappa}{\ln(z/z_0) - \psi_m(z/L)}\right]^2 \qquad z_0 = z_0^{\text{smooth}} + z_0^{\text{rough}}$$

1. Wind Speed Dependent Formulation

$$z_0^{\text{rough}} = \alpha \frac{u_*^2}{g} \qquad \alpha = f_1(U_{10N}) = mU_{10N} + b$$

Charnock coefficient

Wind-wave equilibrium, wind-aligned waves

2. Wave-Based Formulation

$$z_o^{rough} = H_s 0.09 \frac{u_*^{\,2}}{C_p}$$

significant wave height

dominant phase speed

Wind-aligned waves, not appropriate for mixed seas

3. Revised Wave-Based Formulation (Sauvage et al. 2023, 2024)

$$z_o^{rough} = H_s 0.09 cos\left(0.45\theta\right)\left(\frac{u_*}{C_p}\right)^{2cos\left(-0.32\theta\right)}$$

wave-wind misaligment

$$z_o^{rough} = H_s 0.09 \frac{u_*^{\,2}}{C_m}$$

spectrally averaged phase speed

Account for sea states with multiple spectral peaks and waves misaligned with the wind.

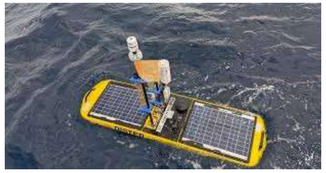
Typhoon Khanun (2023)

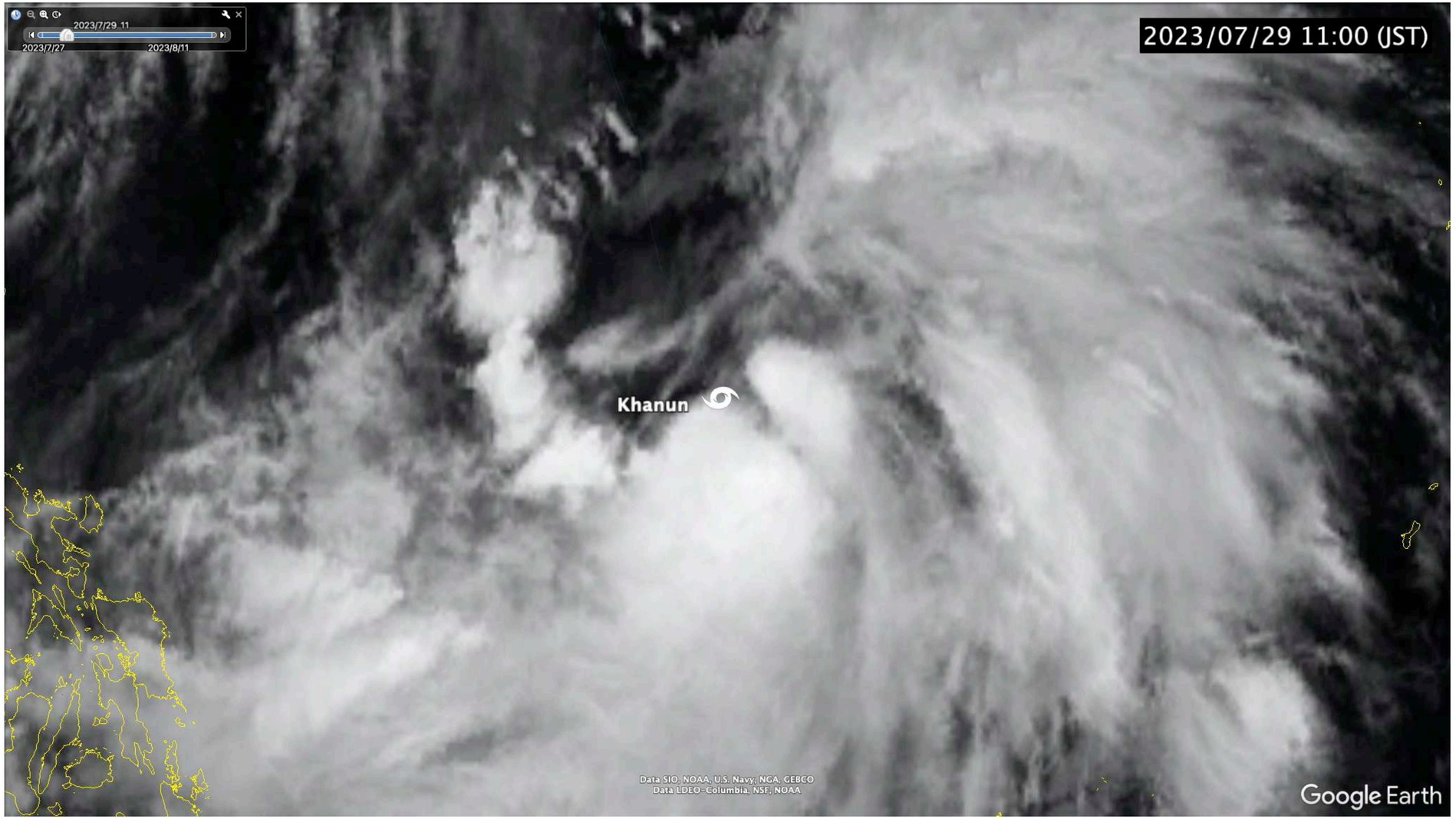
5x



(2 survived and neared the storm center twice)

2x

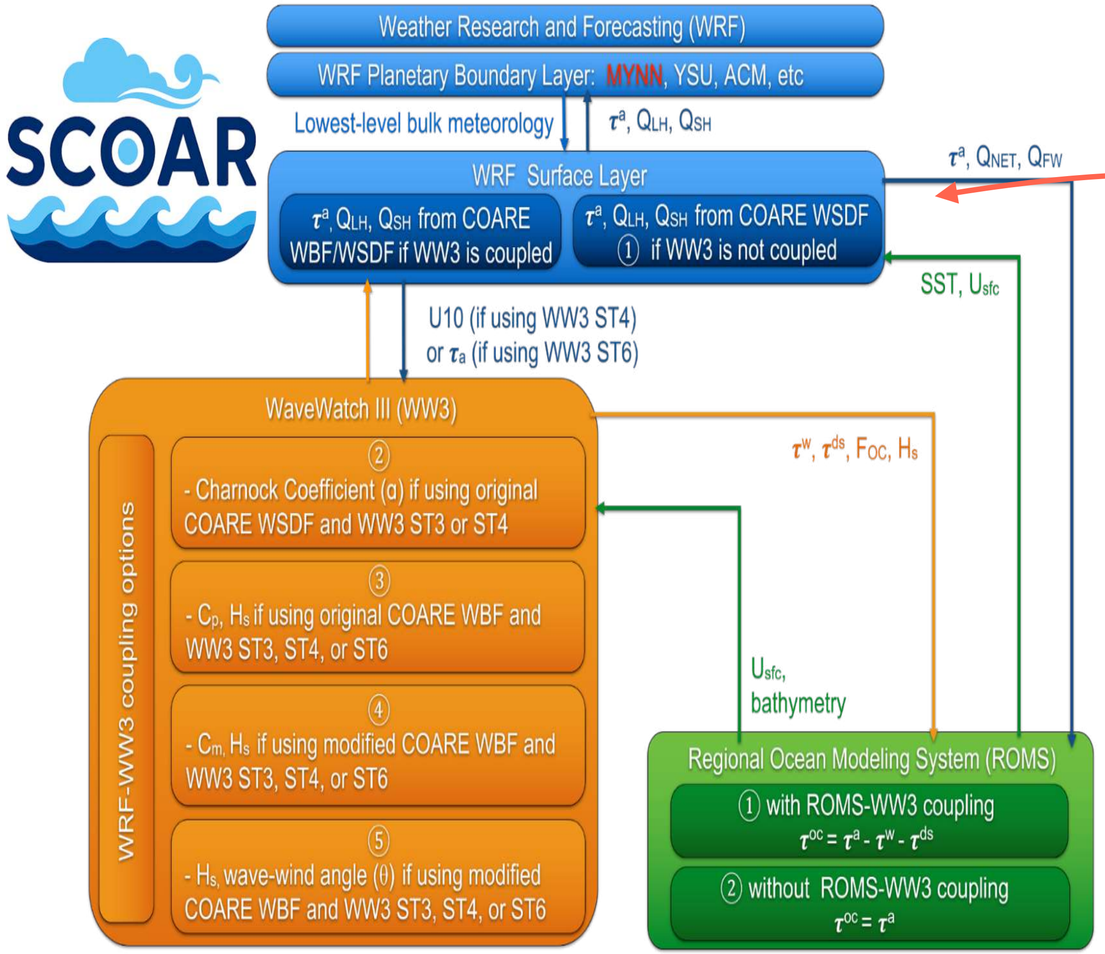




Courtesy: Prof. Satoshi Mitarai (OIST)

SCOAR Regional Coupled Modeling for Typhoon Khanun (2023)

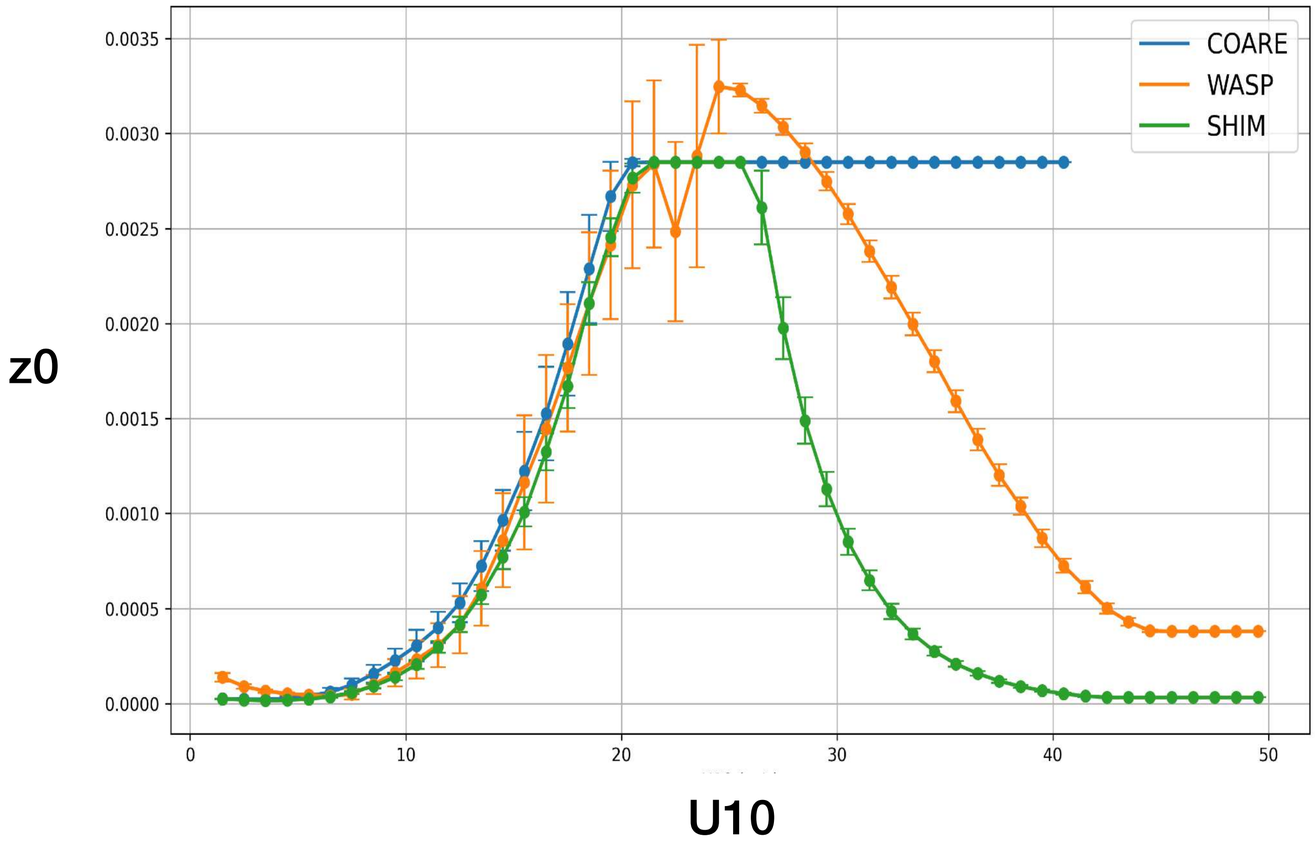
2.5km SCOAR WRF-ROMS-WW3 fully-coupled simulations using 3 different drag coefficient rolloff scenarios



Seo et al. (2017, 2016) Sauvage et al. (2023, 2024)

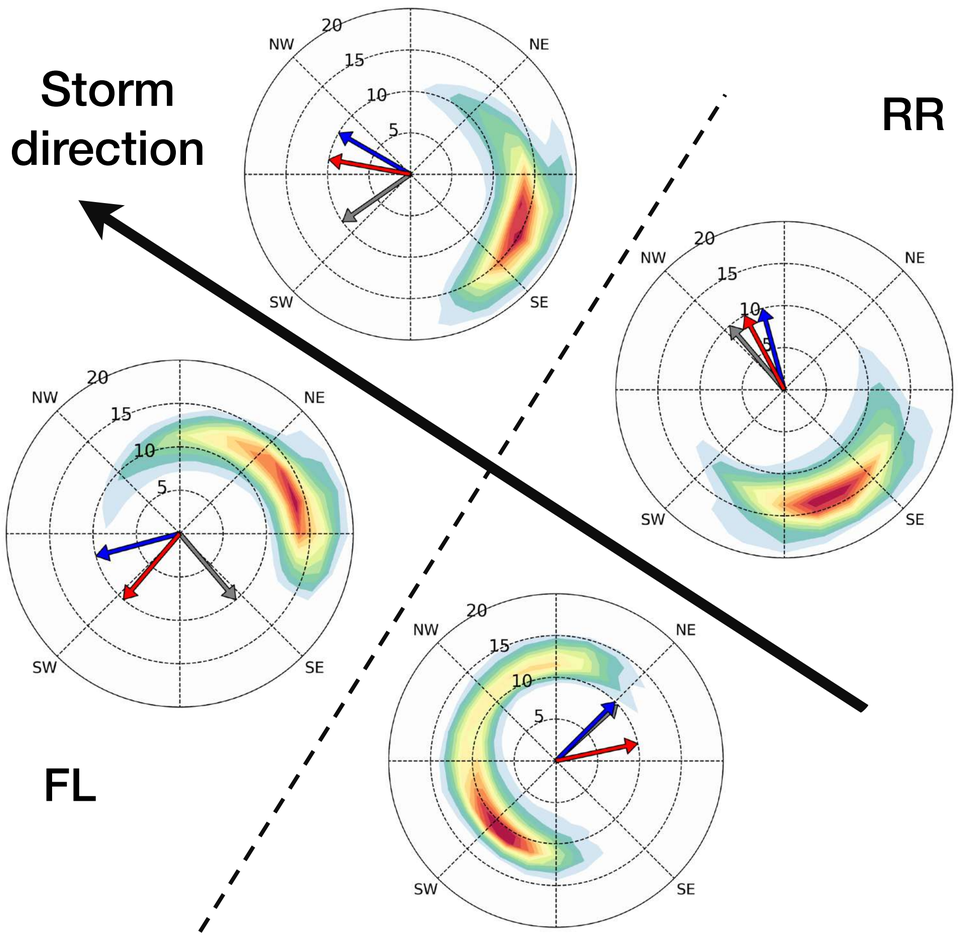
- COARE (Edson et al. 2013; Sauvage et al. 2023)
- WASP (Bouin et al. 2024)
- SHIM (Shimura et al. 2024)

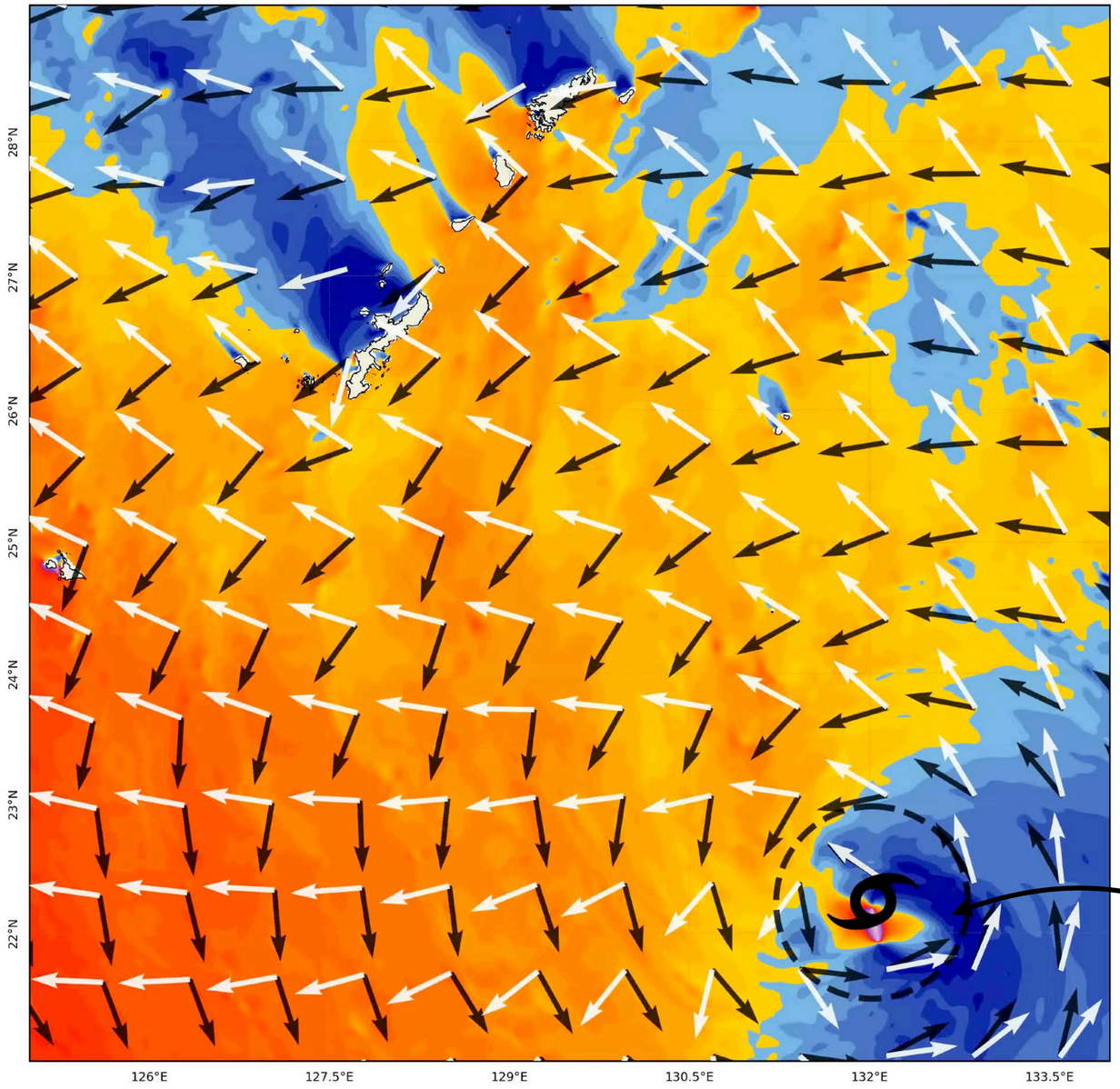
Roughness length (z_o)



Directional Wave-Wind Coupling under Khanun

↖ mean wave direction
↖ peak wave direction
↖ wind direction





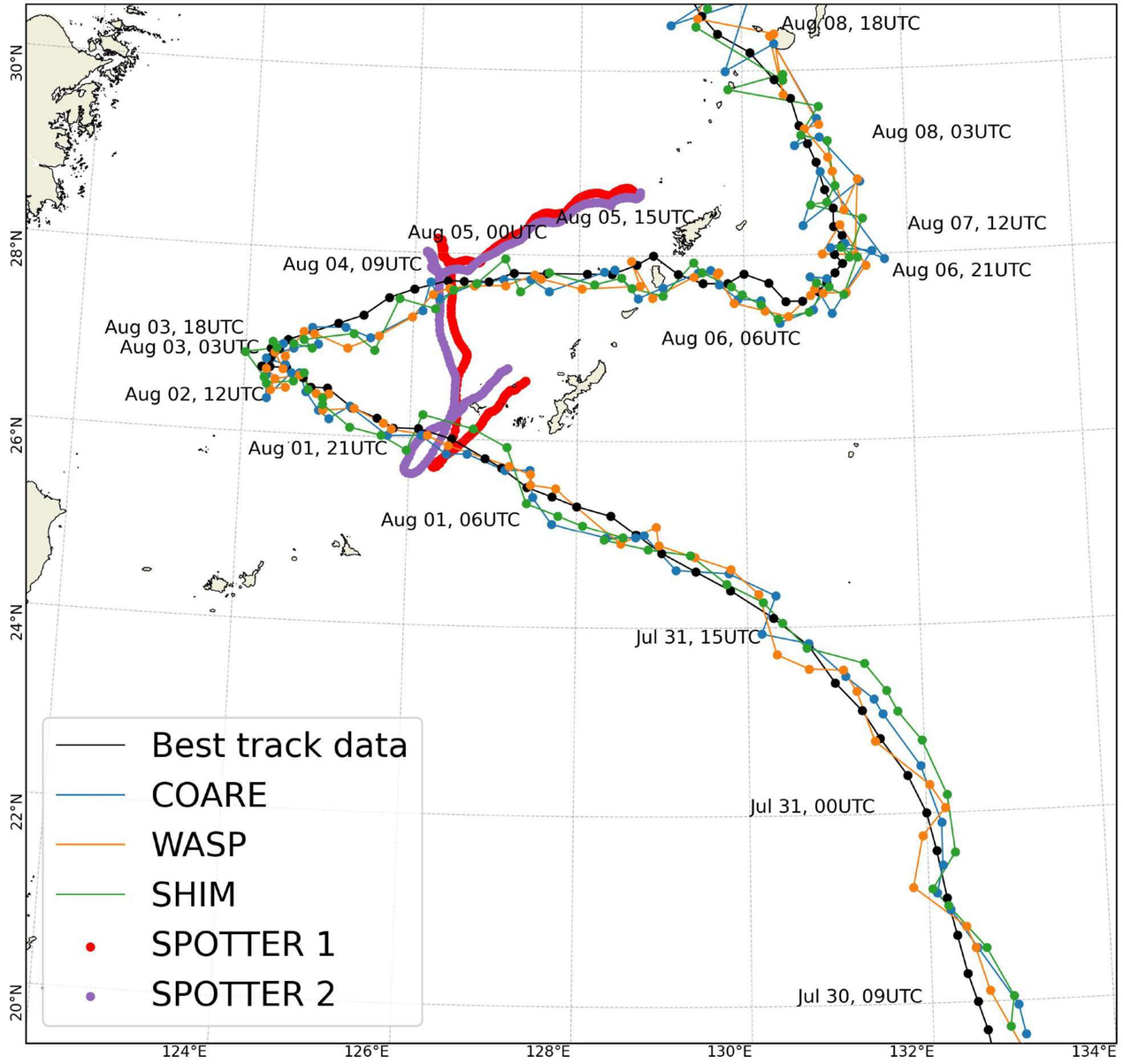
↖ mean wave direction
↖ wind direction

θ (°)
180
135
90
45
0
Opposed
Crossing
Aligned

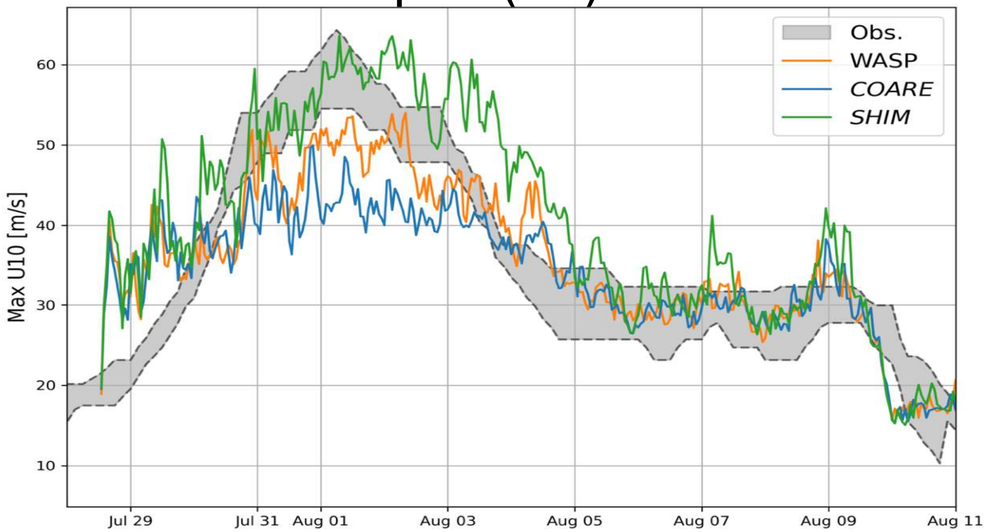
Radius Max. Wind

July 31—Aug 2

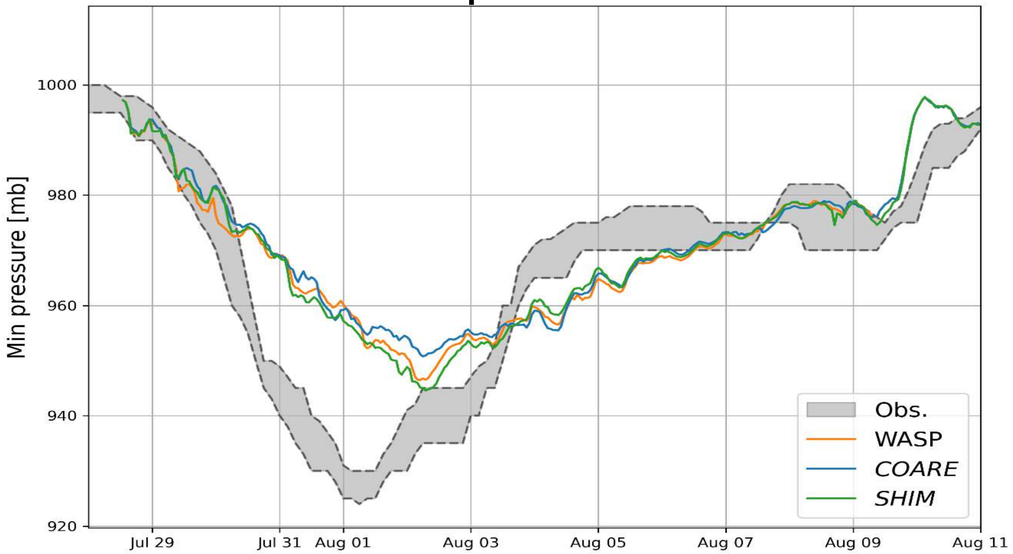
Typhoon Khanun (2023): Storm Intensity



Maximum wind speed (m/s)

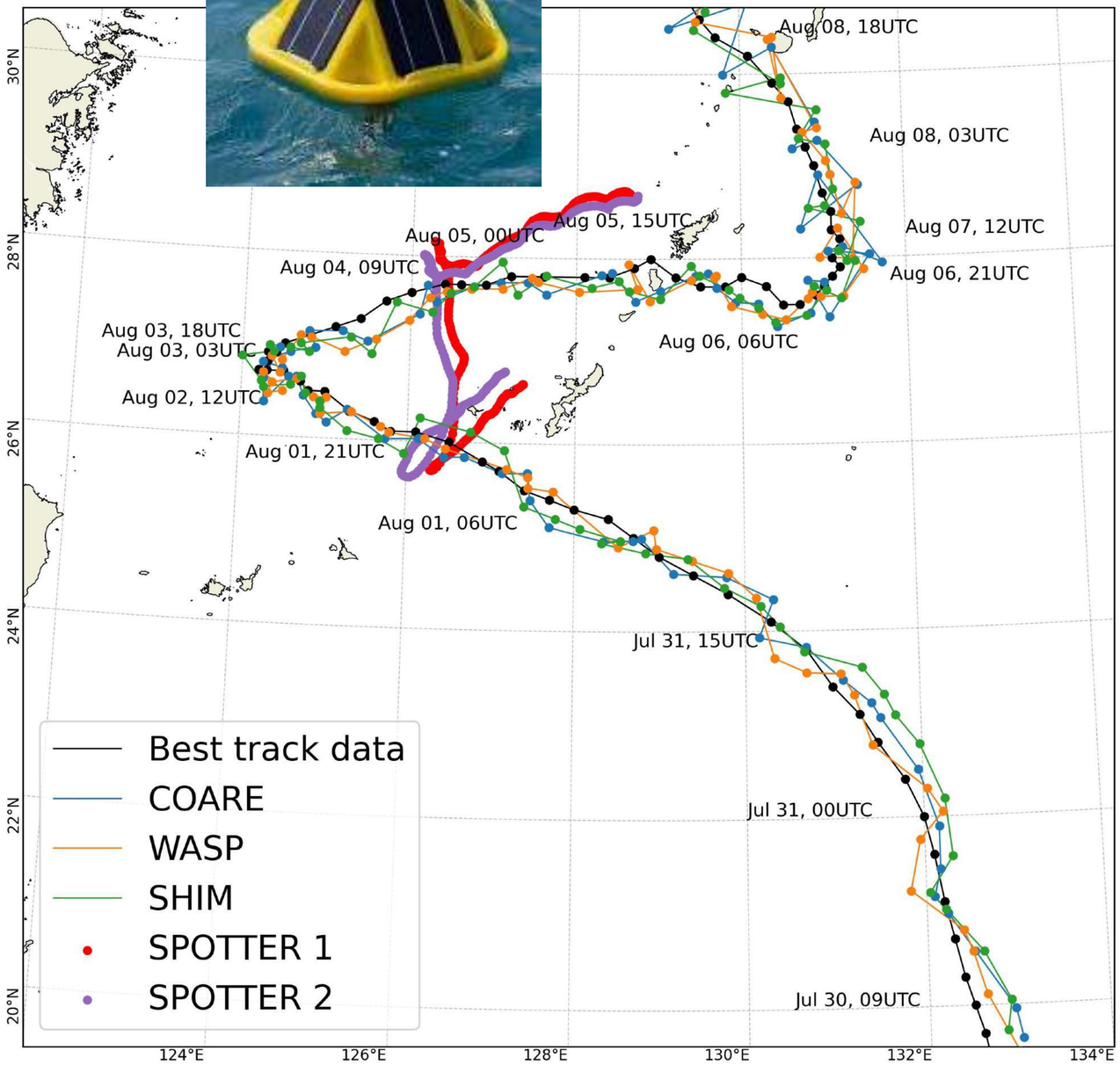


Minimum Sea level pressure

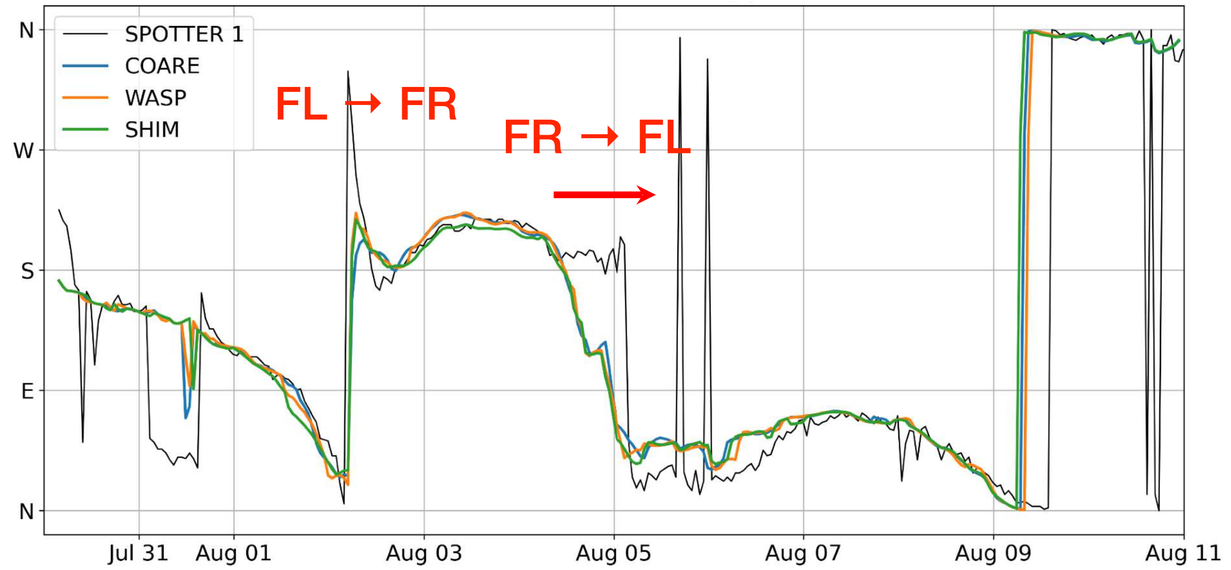


Typhoon Khanun (2023): Sea states

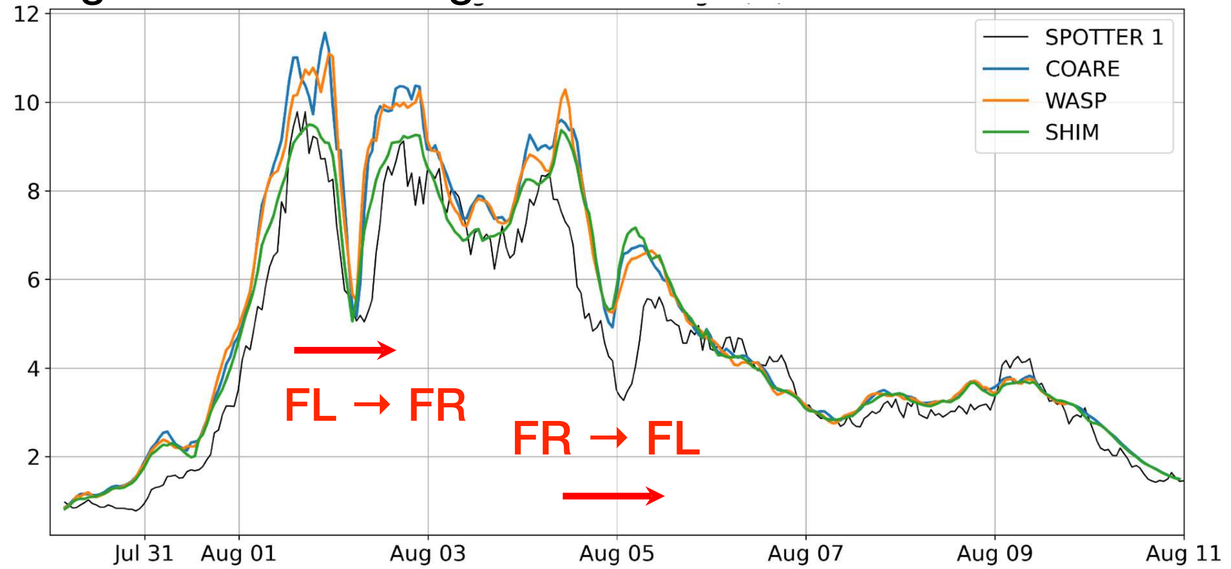




Peak wave direction

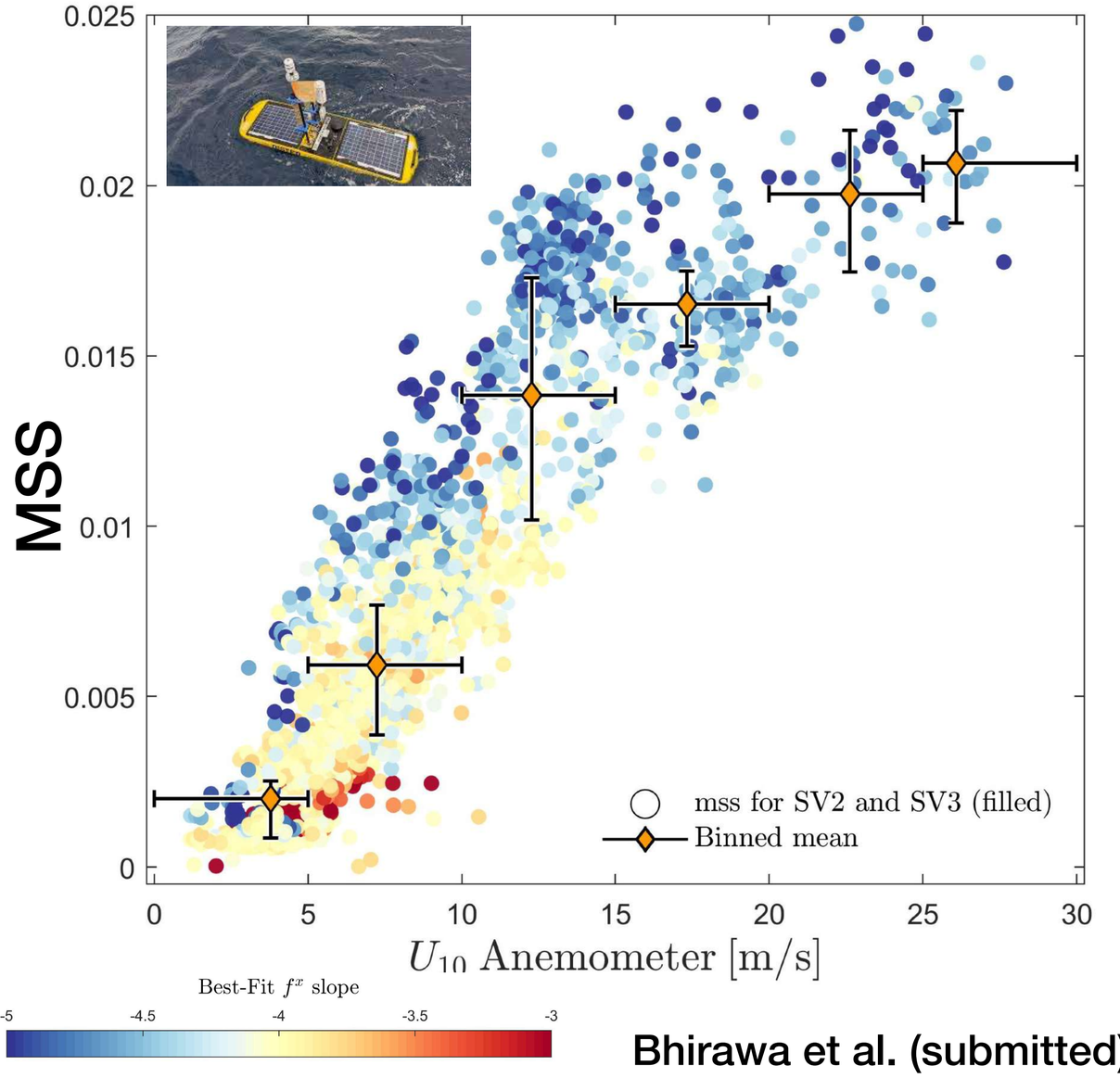


Significant wave height



MSS Saturation

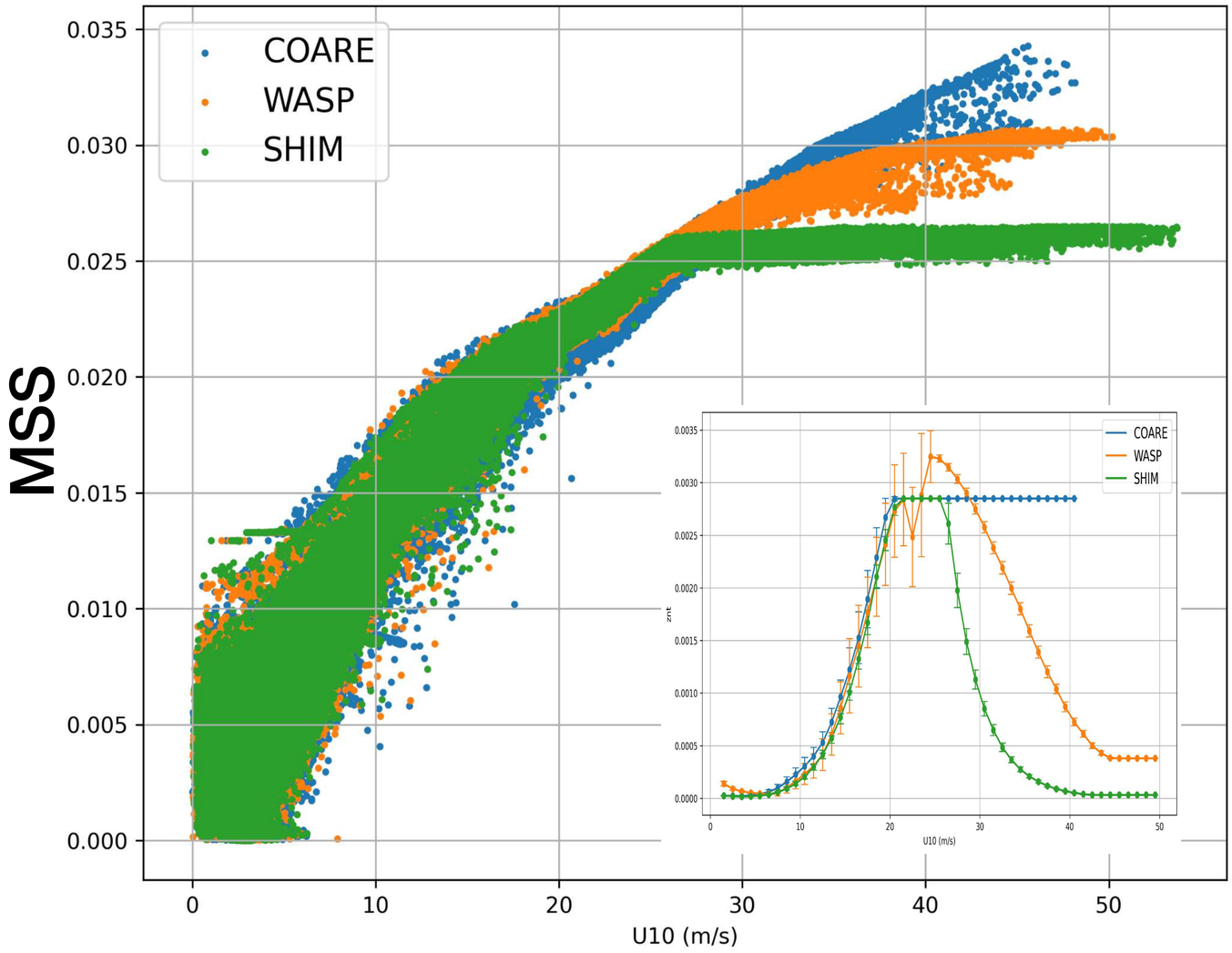
Waveglider-based estimates



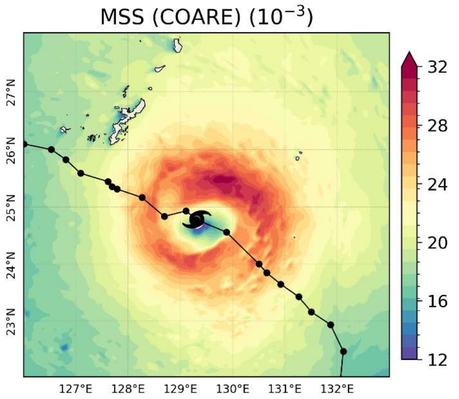
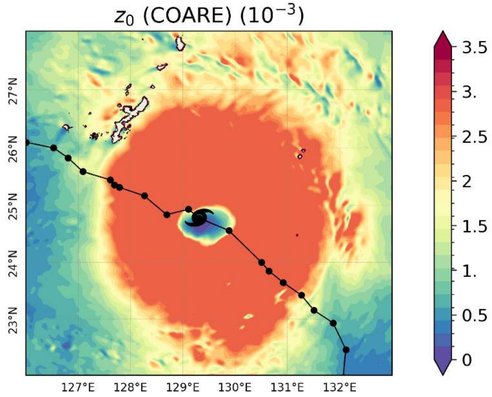
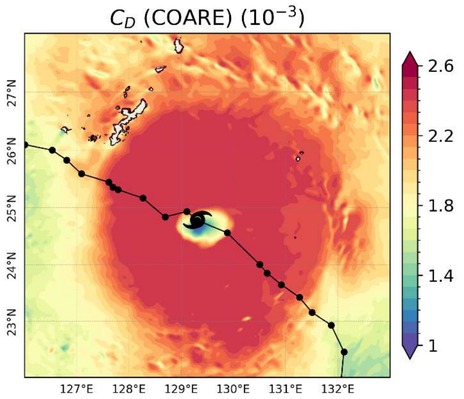
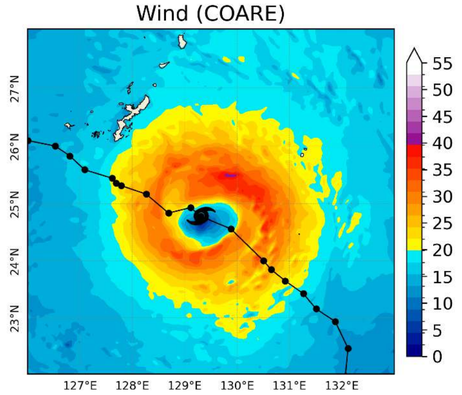
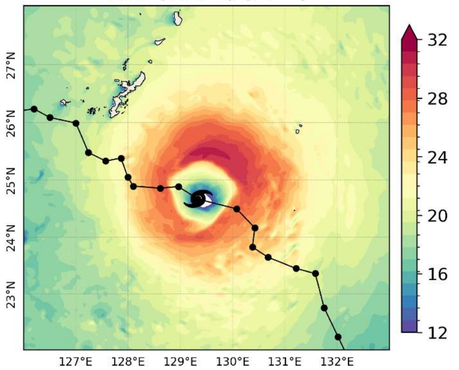
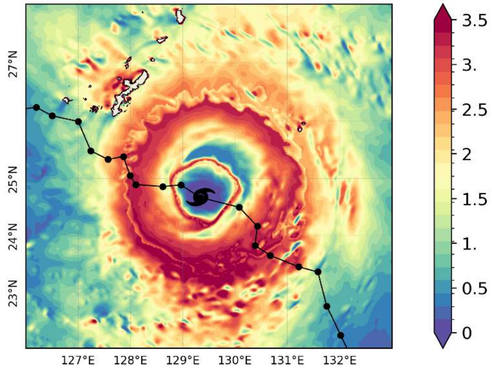
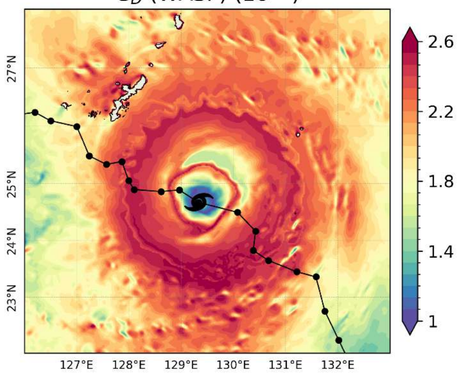
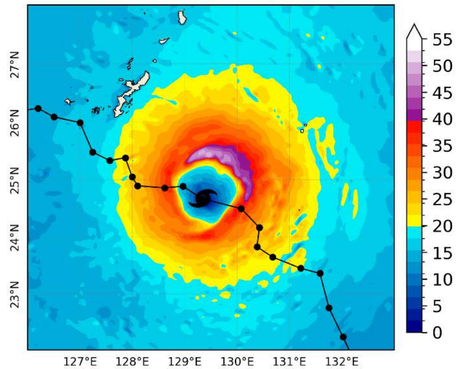
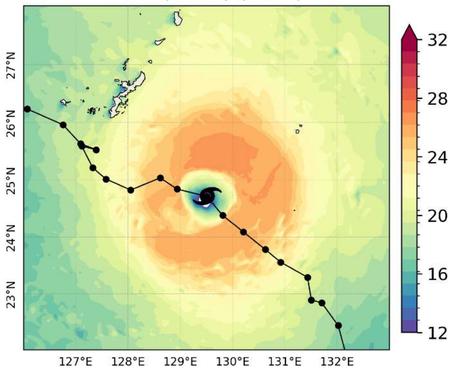
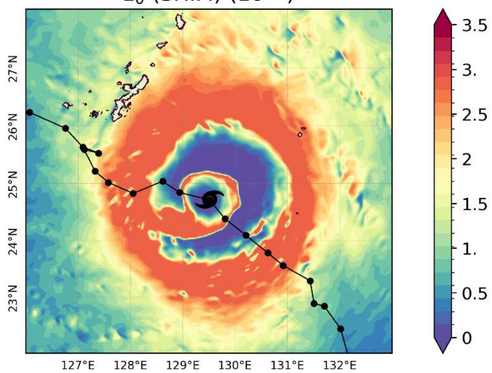
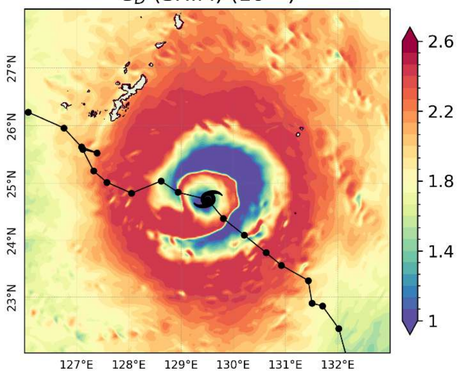
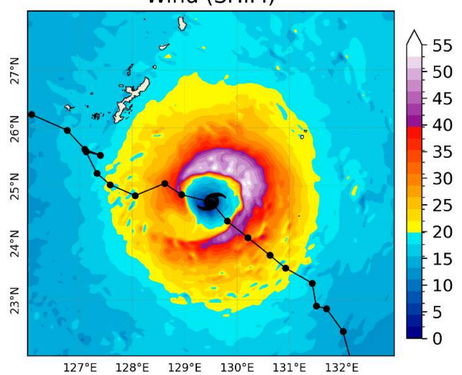
Bhirawa et al. (submitted)

0 m/s

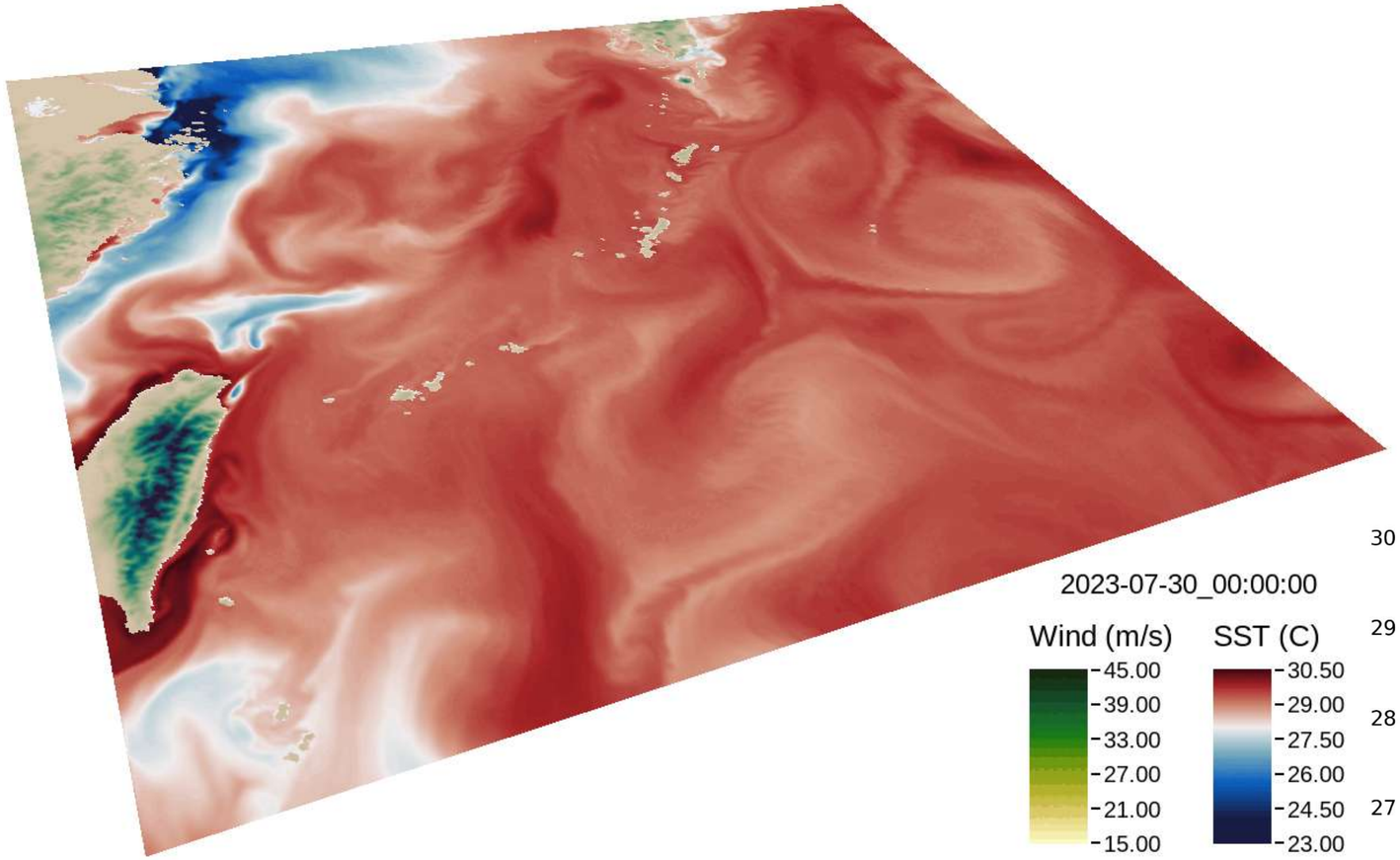
30 m/s



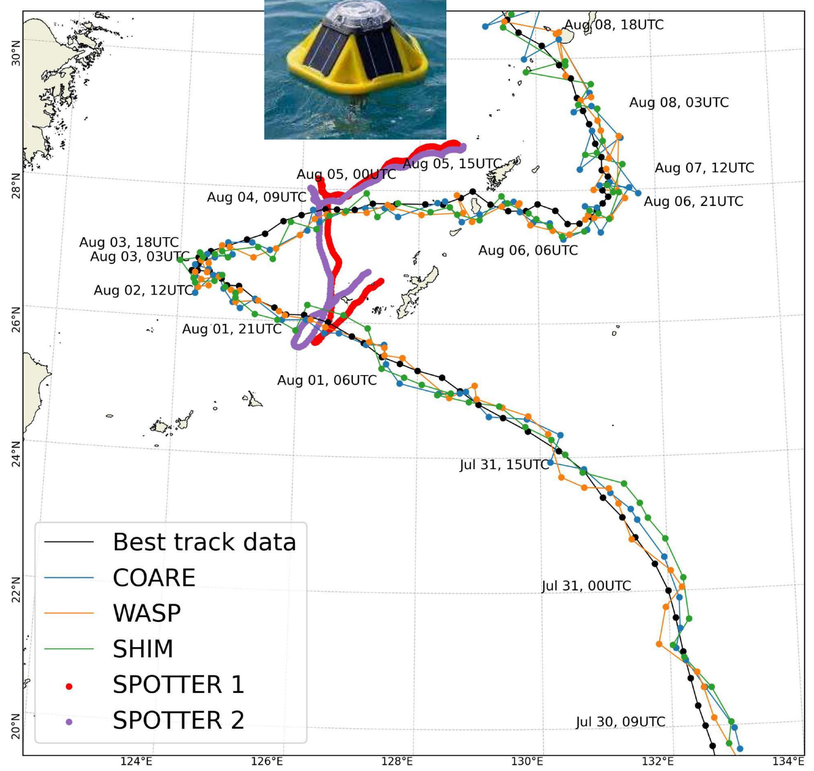
Drag Coefficients, Wave Energies, and Air-Sea Momentum Exchanges

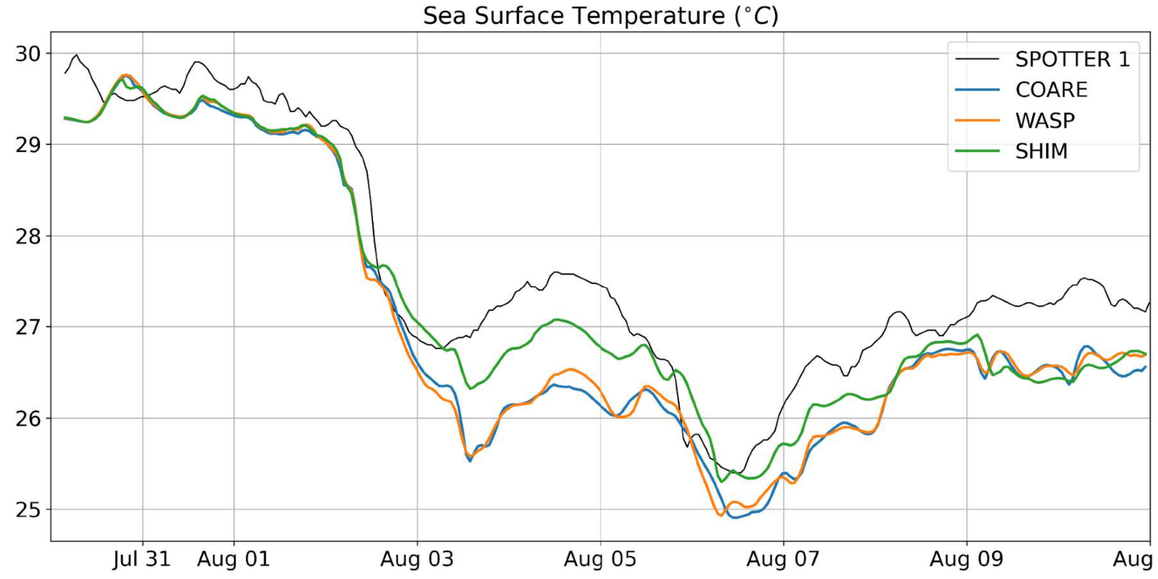
	MSS	z_0	C_D	U_{10}
COARE				
WASP				
SHIM				

Typhoon Khanun (2023) Cold Wakes





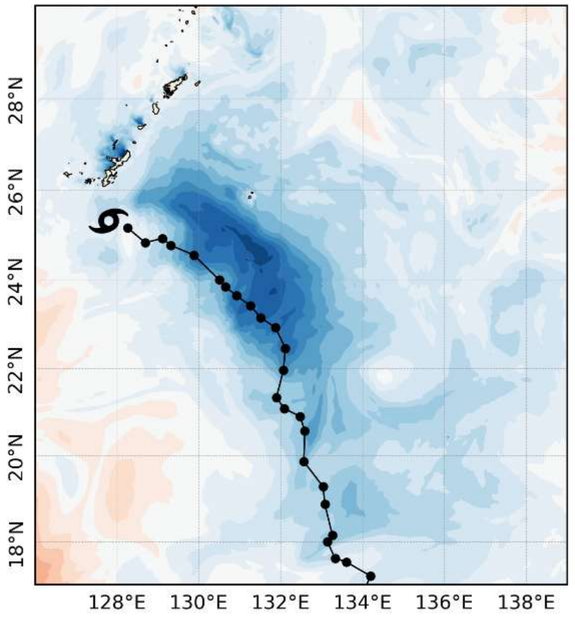




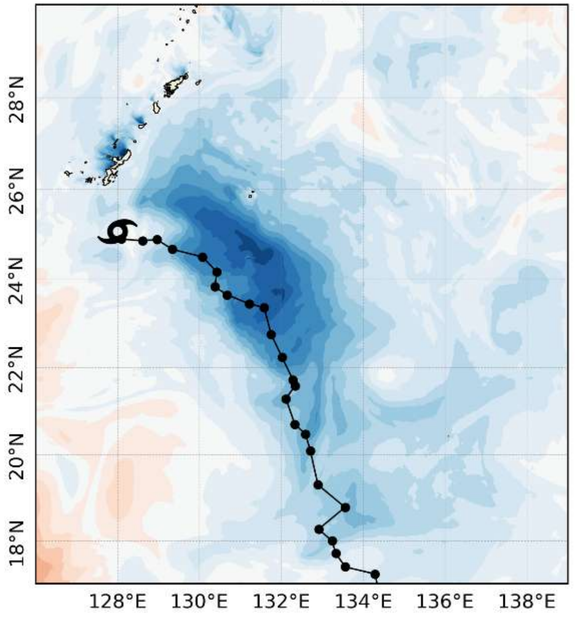
Typhoon Khanun (2023) Cold Wakes

ΔSST (8/1/26 - prestorm)

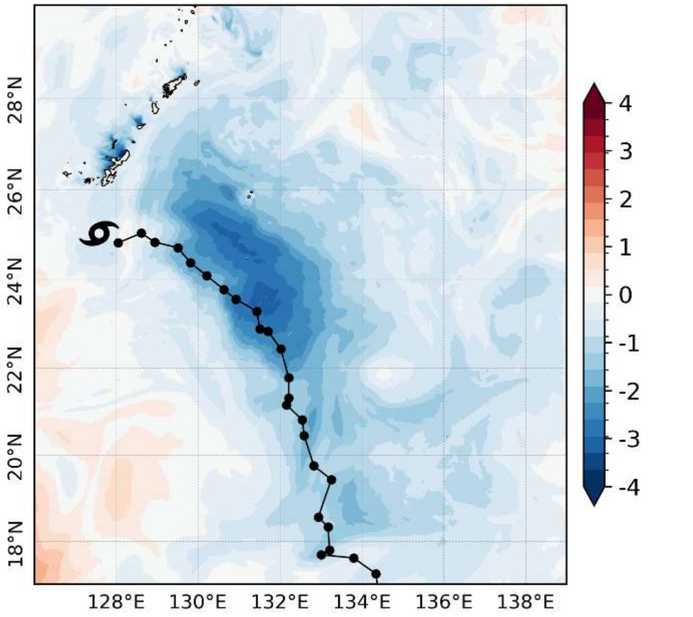
COARE



WASP

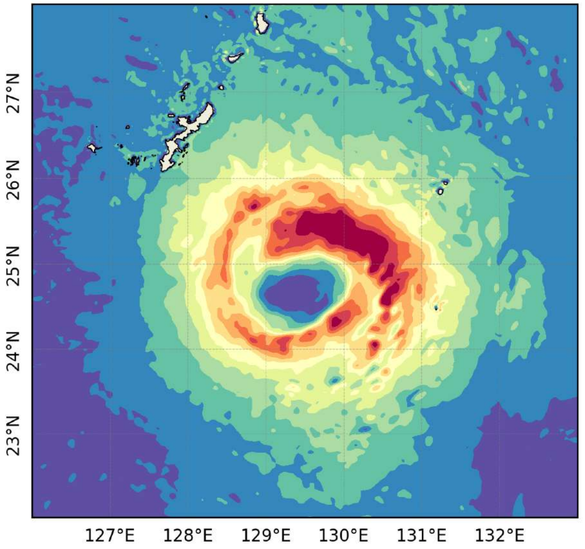


SHIM

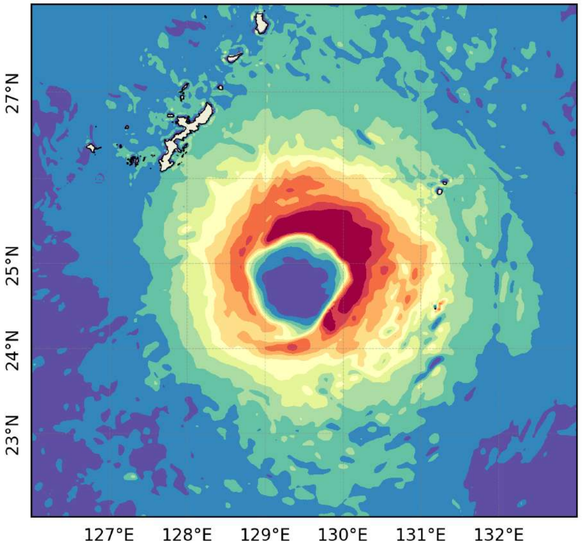


Near-surface TKE

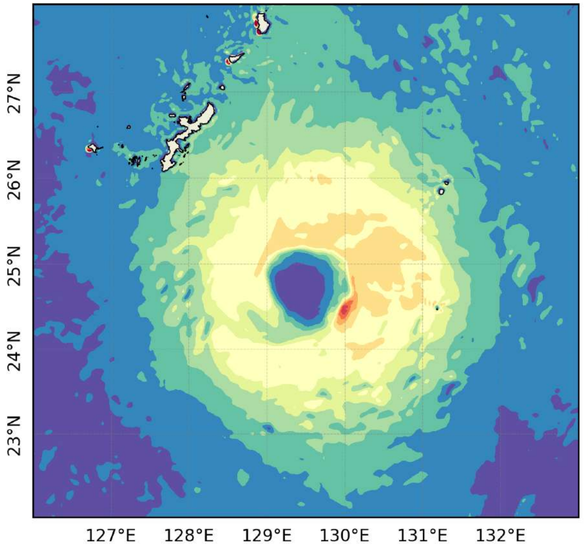
TKE (COARE)

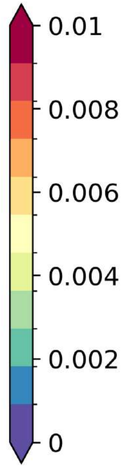


TKE (WASP)



TKE (SHIM)

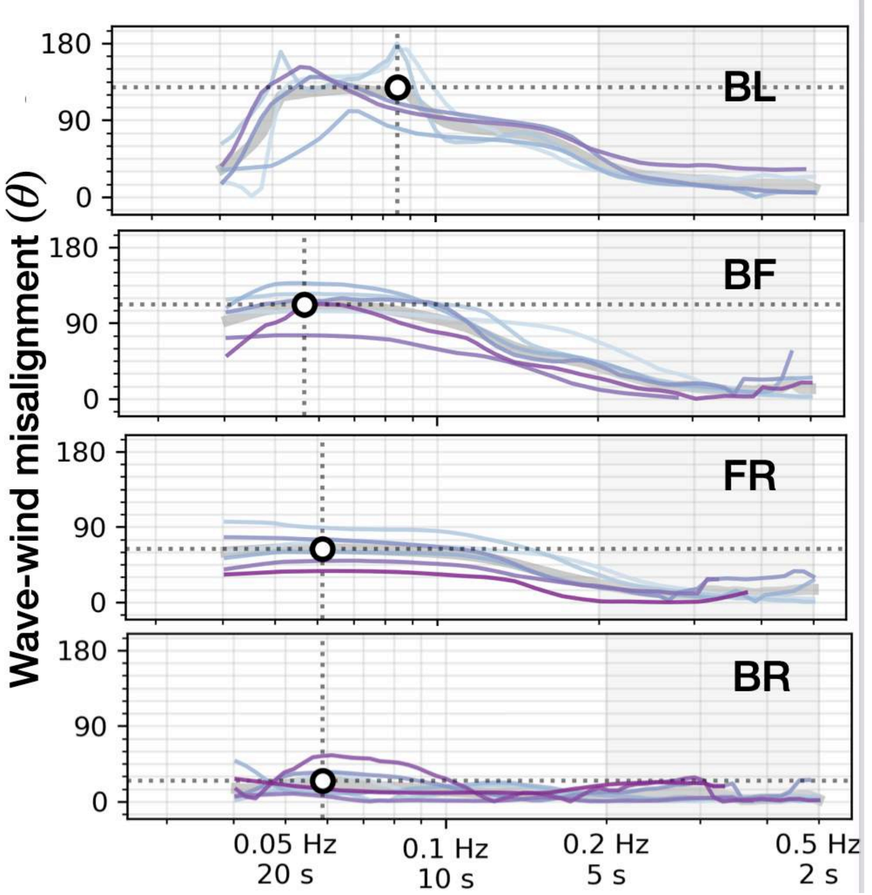




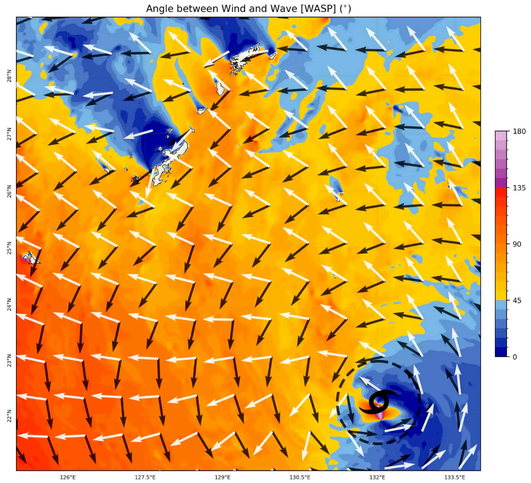
Conclusion

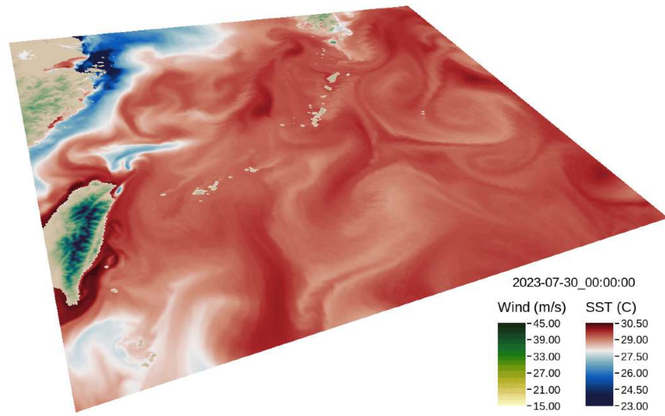
- Observed strong quadrant (and storm radius)-dependent spatial variability in MSS, wind-wave misalignment, and drag coefficient.

- The observationally constrained formulation for z0 of Shimura et al. (2024) produces the best agreement, with pronounced quadrant-dependent reductions in C_D and MSS.



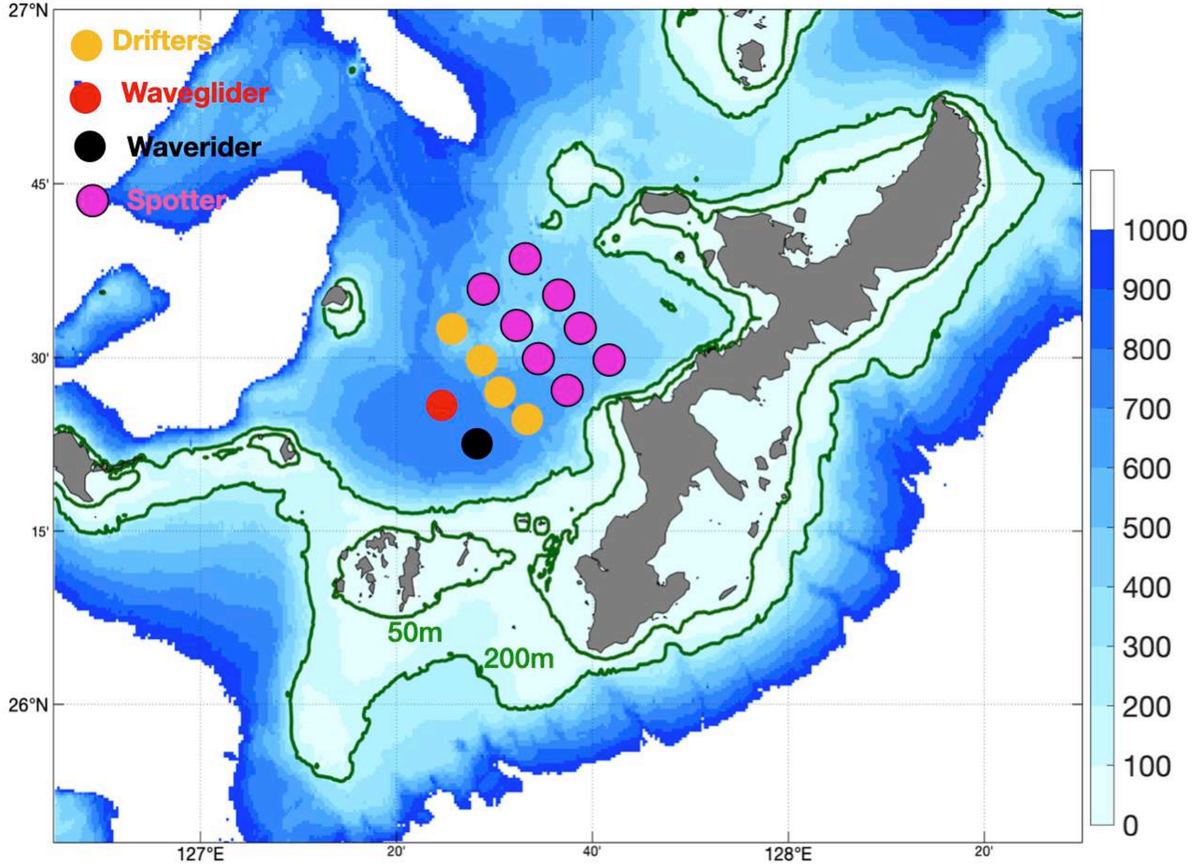
- The choice of the z0 formulation in high winds has a substantial impact on storm intensity.
- In this study, Lower z0/C_D leads to
  → higher winds, weaker cold wake, weakly increased enthalpy fluxes (despite reduced C_K).





Planned Observations : August 2026

Collaboration: UHM & OIST +SIO & WHOI



1x UH PacIOOS wave buoy



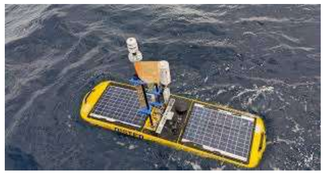
9x OIST
2x Kyoto U
4x UH

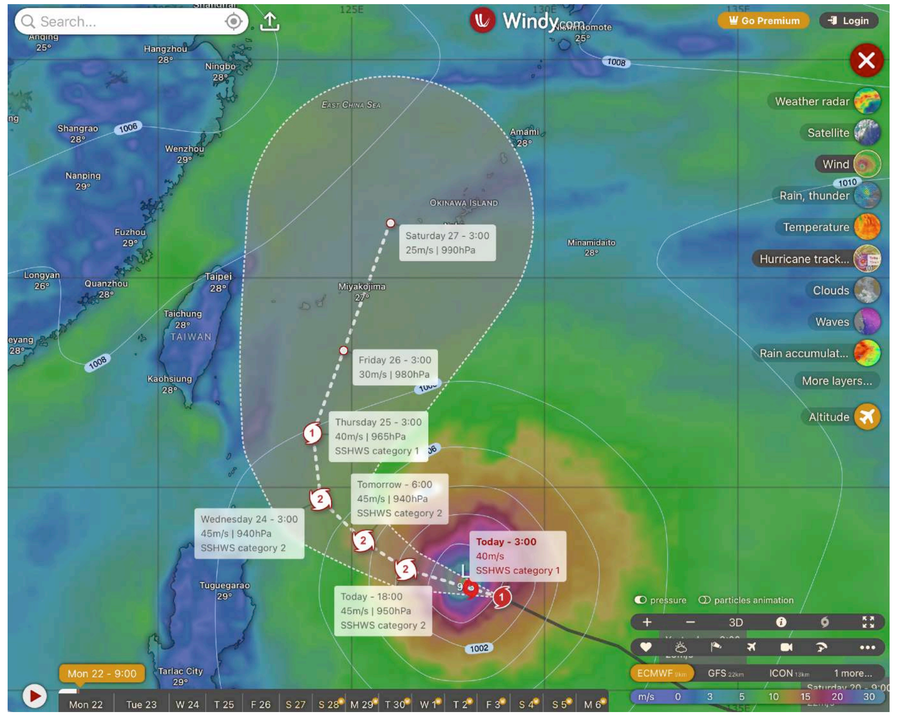


x20 CODE Drifters



1x (SIO-OIST)

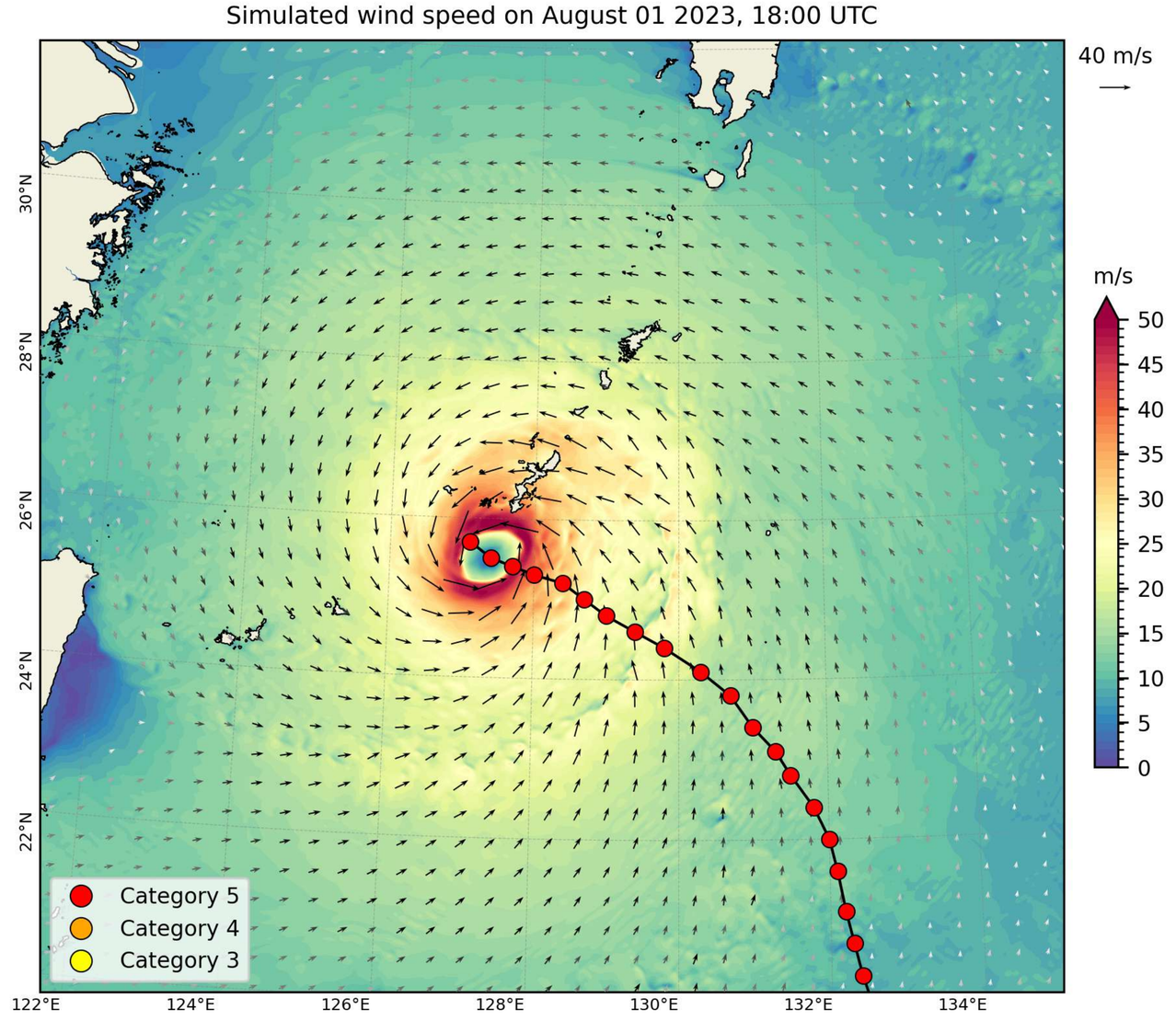




Typhoon Mekkhala

Typhoon Khanun (2023): Simulation

U10



Hs

